LITERATURE SURVEY REPORT

Title: Student Feedback Sentiment Analysis System for Distance Education using ARM with K-Means Clustering

1. Introduction

Sentiment Analysis is a key application of Natural Language Processing (NLP) used to extract opinions, emotions, and attitudes from textual data. In the context of education, analyzing student feedback helps institutions understand teaching effectiveness and improve learning outcomes.

With the increasing use of online platforms and distance education, large volumes of student feedback are generated. However, analyzing this data manually is time-consuming and inefficient. Additionally, challenges such as fake reviews and difficulty in identifying implicit and explicit opinions reduce the effectiveness of traditional systems.

This paper presents a sentiment analysis system that addresses these challenges using advanced techniques such as Association Rule Mining (ARM), K-Means clustering, and lexicon-based classification. This system focuses on sentiment classification, feature extraction, and opinion summarization.

2. Problem Statement

The existing sentiment analysis systems face several limitations:

- Presence of **fake or spam reviews**, which reduces data reliability
- Difficulty in identifying **implicit and explicit features** in feedback
- Lower accuracy in traditional classification techniques
- Lack of proper feature extraction methods

To overcome these issues, the proposed system integrates fake review detection, clustering techniques, and sentiment classification to improve accuracy and efficiency.

3. Methodology

The proposed system architecture consists of the following stages

3.1 Data Collection

The dataset is obtained from the Kaggle platform, consisting of student feedback related to courses, faculty, and distance education systems.

3.2 Fake Review Detection

To ensure data quality, fake reviews are identified and removed using several indicators:

- Review Relevancy Rate
- Content Length
- Review Content Similarity

A **spamicity score** is calculated using these factors. Reviews exceeding a predefined threshold are considered fake and removed.

3.3 Data Preprocessing

The collected data is cleaned and prepared for analysis using:

- Stopword removal
- Tokenization
- Part-of-Speech (POS) tagging

This step reduces noise and improves the quality of input data.

3.4 Feature Extraction

The system uses **Association Rule Mining (ARM) combined with K-Means clustering**:

- ARM identifies relationships between words and features
- K-Means groups similar features together

K-Means clustering groups similar features based on similarity, which helps in better identification of related aspects.

- Explicit features (directly mentioned)
- Implicit features (indirectly expressed)

3.5 Sentiment Classification

A Lexicon-based method is used for sentiment classification:

- Uses a predefined dictionary of words with sentiment scores
- Classifies feedback into:
    - Positive
    - **Negative**

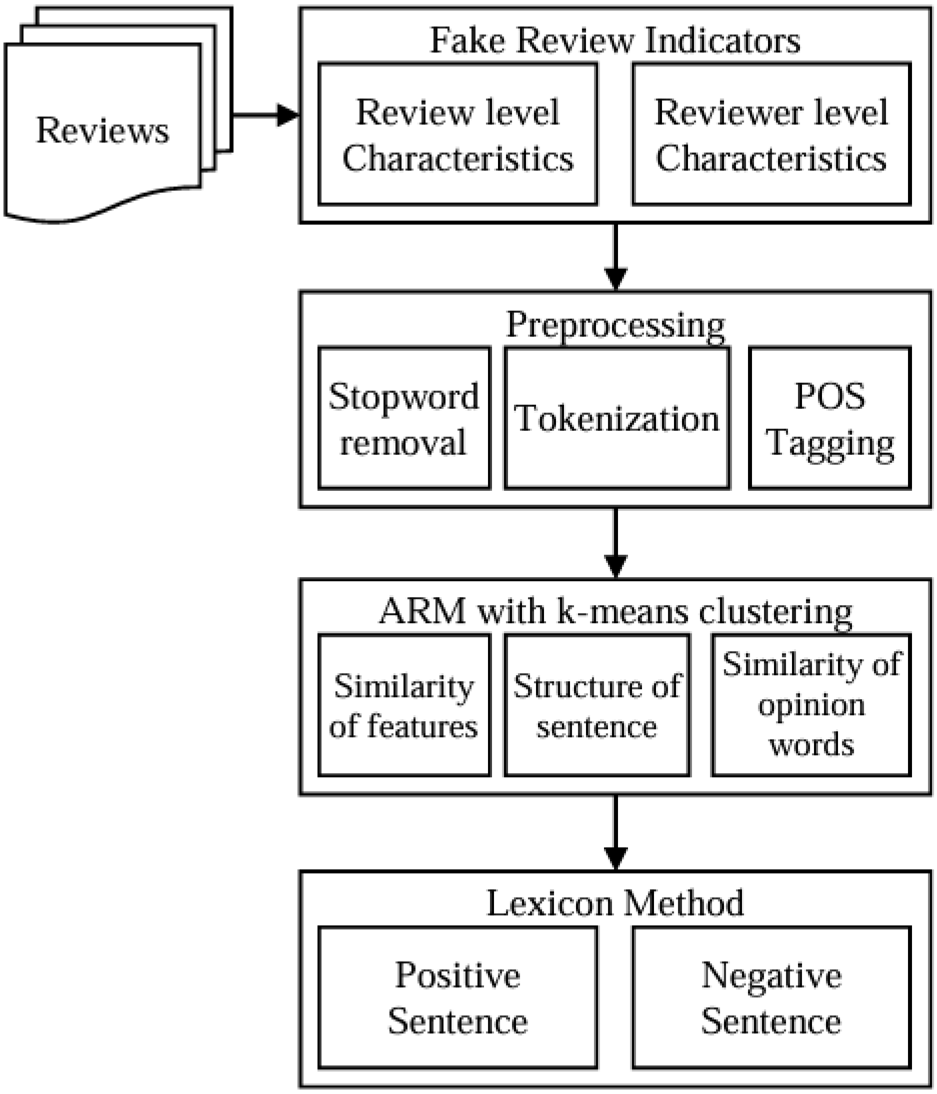

Fig.1. Architecture of proposed system

4. Evaluation Metrics

The system performance is evaluated using standard metrics:

- **Precision**: Measures the correctness of predictions
- **Recall**: Measures the ability to identify relevant results
- **F-Measure**: Harmonic mean of precision and recall

These metrics help in evaluating the accuracy and effectiveness of the proposed system.

5. Results and Discussion

The proposed system shows improved performance compared to existing methods:

- Higher **precision, recall, and F-measure values**
- Effective detection and removal of fake reviews
- Better identification of implicit and explicit features
- Performance comparable to human evaluation

The integration of ARM with K-Means clustering significantly enhances feature extraction and classification accuracy. The system achieves improved classification performance due to better feature extraction and removal of noisy data.

6. Conclusion

The paper presents an efficient student feedback sentiment analysis system that overcomes the limitations of traditional methods. By combining fake review detection, clustering techniques, and lexicon-based sentiment analysis, the system achieves improved accuracy and reliability.

The proposed approach is effective in analyzing student feedback in distance education systems and can be extended to other domains for opinion mining and sentiment analysis tasks.

7. References

Harshini, G.N., and Gobi, N.,
"Student Feedback Sentiment Analysis System for Distance Education using ARM with K-Means Clustering,"
ICTACT Journal on Soft Computing, Vol. 10, Issue 3, April 2020.